

Literature Survey on Smart Digital Complaint Management and Public Transparency System (UrbanFix)

A Zero-Registration, OTP-Verified Civic Grievance Redressal Platform with Severity-Aware Triage and Proposed AI-Assisted Photo Validation

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1. Introduction

Urban local bodies in India receive a large volume of everyday civic complaints—potholes, garbage accumulation, water leakage, faulty streetlights, and similar infrastructure issues. Citizens today report these problems through phone hotlines, in-person visits to municipal offices, or informal posts on social media. None of these channels give the citizen a tracking ID, a public record of what was reported, or any visible sense of how urgent their complaint is compared to others. This fragmentation is the starting point for our project, UrbanFix, and for most of the research reviewed in this survey.

Over the last two years, a wave of student and academic research projects (reviewed as [1]–[7] in this survey) has proposed AI-assisted civic grievance platforms that try to automate one or more parts of this pipeline: classifying complaints into the right department, estimating urgency, detecting duplicate reports, or verifying photo evidence. This survey studies seven such systems, compares their approaches with the approach taken in UrbanFix, and identifies what gap in the existing literature UrbanFix is trying to address.

2. Background

Civic grievance redressal sits inside the broader field of e-governance and smart-city computing. A grievance system typically needs to do four things: (a) let a citizen submit a complaint with enough detail to act on, (b) route that complaint to the correct department, (c) decide how urgently it needs attention, and (d) let the citizen (and ideally the public) see what happened to it. Government portals such as CPGRAMS and city-specific apps such as the BBMP Sahaaya App already exist for step (a), but the papers reviewed here repeatedly point out that these portals fall short on steps (b)–(d): routing is manual, urgency is not scored, and status tracking is weak or absent [1], [2].

Two technical trends run through the reviewed literature. The first is NLP-based text classification using TF-IDF, sentiment analysis, or transformer models (zero-shot classifiers, DistilBERT, BART) to automatically read a complaint's text and decide which department should handle it and how urgent it is. The second is computer-vision-based image handling using cloud storage, TensorFlow Lite, CNNs, or YOLO object detectors either to store photo evidence, moderate inappropriate images, classify what is in the photo, or (as UrbanFix proposes) verify that the photo actually matches the reported defect category or not.

3. Problem Statement

Civic issue reporting today is fragmented, untracked, and largely invisible to the public. A citizen who calls a hotline, visits a municipal office, or posts on social media has no tracking ID, no assurance that the complaint

has been logged anywhere permanent, no sign of how urgent their issue is relative to others, and no photo-verification step to confirm the issue is genuine. Individually, each of these gaps is an inconvenience; together, they add up to a systemic breakdown in civic accountability and public trust ; which is the problem UrbanFix is built to address.

4. Overview of the Proposed Project (UrbanFix)

UrbanFix is a web-based, account less civic complaint reporting and public transparency platform. A citizen picks one of 10 fixed infrastructure categories (potholes, garbage, water leakage, streetlights, and similar), answers a short weighted questionnaire that produces a live, explainable priority score, uploads photo evidence and a description, and verifies their submission with a one-time email OTP ; no account or password is ever created. Once verified, the system issues a permanent tracking ID and the complaint immediately appears in a public registry, visible to any visitor, with its full status history (not just to the person who filed it).

On the backend, an admin console (protected with JWT authentication) lets municipal staff move complaints through a status workflow and issues a PDF receipt to the citizen. The current implementation covers all of this end-to-end using a React/Vite frontend, an Express/Node.js REST API, and MongoDB (with a TTL-indexed OTP collection so codes expire automatically). One further layer is architecturally designed but not yet built: a per-category YOLO object-detection model that would check whether an uploaded photo genuinely matches its claimed category, returning a confidence score and an annotated image, with low-confidence cases automatically routed to a human admin review queue rather than auto-published. This keeps a human in the loop even once the AI layer is added.

5. Research Questions

Based on the project and the seven papers reviewed, this survey is organised around three research questions:

- RQ1: What approaches have existing AI-assisted civic grievance systems used for complaint classification, urgency scoring, and photo/media verification, and what limitations do they report or exhibit?
- RQ2: To what extent do current systems provide a single, publicly visible complaint registry (with full status history open to any citizen) combined with an explainable, severity-aware priority score rather than an admin-only dashboard and an opaque ML urgency label?
- RQ3: How can a zero-registration, OTP-verified submission flow combined with a human-in-the-loop, category-specific AI photo-validation layer address the gaps found in RQ1 and RQ2, and is this achievable with a lean, open-source stack?

Keyword Matrix

Concept	Main Keywords	Related Keywords
Problem	civic complaint, grievance redressal	untracked complaints, public accountability, infrastructure issue reporting
Technology	AI, NLP, machine learning	zero-shot classification, sentiment analysis, TF-IDF, object detection (YOLO)

Concept	Main Keywords	Related Keywords
Application	municipal governance, smart city	e-governance, urban infrastructure management
Users	citizen, municipal officer	resident, administrator, field staff
Proposed Solution	OTP verification, public registry	priority scoring, photo validation, account-less reporting
Domain	civic technology	urban informatics, digital governance

Boolean Search Strings

- ("civic complaint" OR "grievance redressal") AND ("machine learning" OR "artificial intelligence") AND "municipal"
- "complaint classification" AND ("NLP" OR "zero-shot" OR "transformer") AND "urban"
- ("public grievance" OR "citizen complaint") AND "geolocation" AND "image"
- "civic issue reporting" AND ("transparency" OR "public registry" OR "audit trail")
- "complaint prioritization" AND ("severity" OR "urgency scoring")
- "YOLO" OR "object detection" AND "complaint verification" AND "civic"
- "e-governance" AND "rural" AND "urban" AND "digital divide"
- "smart city" AND "grievance" AND "duplicate detection"

6. Review of Existing Research

Infra-Alert — A Web-Based Public Grievance Redressal System with Image and Geolocation Support [1]

What it is about: Infra-Alert is a web-based, crowdsourced complaint platform (HTML/CSS/JS front end, PHP/MySQL back end) built to modernise reporting of infrastructure problems such as potholes, faulty streetlights, garbage, and electricity faults.

Problem addressed: It targets the same fragmentation problem as UrbanFix: slow, bureaucratic complaint handling with no real-time tracking, poor geolocation accuracy, and no cloud-based image evidence in existing platforms like CPGRAMS and the BBMP Sahaaya App.

Objective: The goal is to give citizens a transparent, real-time complaint pipeline with automated location tagging and image evidence.

Method/Approach: Complaints are submitted through a dynamic form, located automatically using the Geolocation API and Reverse Geocoding, and stored with images in Cloudinary; authorities view, resolve, or reject issues via an authority dashboard and upload "before/after" resolution photos.

Technology/Algorithms: HTML/CSS/JavaScript frontend, PHP backend, MySQL database, Google Maps Geolocation API, Reverse Geocoding, Cloudinary cloud image storage.

Dataset/Experiment: No formal ML dataset — the system was demonstrated through a working prototype with real user-submitted complaint records shown in a comparison table against CPGRAMS, I Change My City, and Public Eye.

Main findings: Infra-Alert supports more features (real-time geolocation, reverse geocoding, cloud image storage, PDF reports, multi-department handling) than the three existing platforms it is compared against, none of which support these together.

Advantages: Strong geolocation accuracy, cloud-based before/after image proof, automated PDF report generation, multi-department routing.

Limitations: Depends on stable internet and GPS accuracy (degrades indoors/in dense areas); AI-driven complaint verification and automated prioritisation are explicitly left as future work, not implemented; the dashboard tracks a citizen's own complaints, not a registry of everyone's complaints.

Research gap: No automated visual verification and no severity/urgency scoring — both are named as future work rather than solved.

Relation to our project: UrbanFix shares the geolocation-and-photo-evidence motivation but takes a different route: rather than PHP/MySQL with a dashboard limited to the submitter, UrbanFix commits to a fully public registry and an explainable priority score from day one, and treats photo *verification* (not just storage) as the extension point.

CitiWatch — AI-Driven Anonymous Civic Intelligence for Transparent and Efficient Municipal Governance [2]

What it is about: CitiWatch is an "Intelligent Anonymous Civic Reporting and Municipal Issue Tracking Platform" that lets citizens file complaints without exposing their identity publicly, while still verifying them.

Problem addressed: It addresses slow manual routing, identity-exposure fears that discourage reporting, and weak tracking transparency in traditional systems.

Objective: To combine anonymity with accountability, and to automate department routing so complaints are not sorted by hand.

Method/Approach: Citizens verify via mobile-number OTP (identity never shown publicly); complaint text is converted into TF-IDF feature vectors and classified by a supervised ML model into the correct department; each complaint gets a unique ID; department dashboards are secured with JWT.

Technology/Algorithms: TF-IDF feature extraction, supervised ML classifier, OTP authentication, JWT-secured dashboards.

Dataset/Experiment: Evaluated on a multi-departmental municipal dataset spanning ten departments (Water Supply, Sanitation, Roads, Electricity, Waste Management, Street Lighting, Public Health, Drainage, Parks & Recreation, Building Maintenance).

Main findings: Reports 96.8% department-classification accuracy, 95.9% precision, 95.4% recall; automated routing cuts allocation time by about 85%; overall resolution efficiency above 93%; OTP authentication succeeds over 97% of the time.

Advantages: Strong quantitative routing-accuracy results; genuinely anonymous-but-accountable design; JWT-secured admin access; measurable time savings over manual routing.

Limitations: Anonymity is mobile-number-based, not account-less — the phone number is still stored and tied to the complaint; no image/photo-evidence handling at all; no public registry — dashboards are department-facing, not open to all citizens; future work explicitly lists geolocation tagging and multilingual support as not yet done.

Research gap: No visual evidence handling of any kind, and no public-facing transparency layer — routing efficiency is solved, public accountability is not.

Relation to our project: CitiWatch's OTP-based verification is conceptually close to UrbanFix's OTP flow, but CitiWatch uses it for anonymity of a persistent phone-linked identity, while UrbanFix uses email OTP purely to verify a one-time submission with no persistent account at all. CitiWatch's strong ML routing numbers are also a capability UrbanFix's current MVP does not yet have (routing is still manual via the admin console).

VidyutSaathi — AI-Driven Complaint Management System [3]

What it is about: An AI framework for prioritising, tracking, and resolving public-service complaints, demonstrated through a case study on electricity-sector complaints called VidyutSaathi.

Problem addressed: Existing complaint systems treat classification, duplicate detection, workload balancing, and escalation as separate, disconnected tasks with no single system doing all of them together.

Objective: To build one integrated pipeline that predicts urgency and resolution time, clusters duplicate complaints, assigns them to departments by workload, and escalates unresolved cases automatically.

Method/Approach: Complaint text is embedded using Sentence-BERT; a multi-task neural network predicts severity class and response/resolution time together; a Spatiotemporal Complaint Clustering Module (an improved DBSCAN using spatial, temporal, and Sentence-BERT-similarity thresholds) detects duplicates/hotspots; a Workload-Aware Dynamic Department Assignment Module balances load using distance and workload cost functions; unresolved complaints escalate up an engineer hierarchy; a separate NLP + CNN + credibility-scoring pipeline flags fake or irrelevant complaints.

Technology/Algorithms: PyTorch multi-task neural network, Sentence-BERT embeddings, improved DBSCAN with PostGIS R-tree indexing, CNN-based image-text consistency checking.

Dataset/Experiment: Over 20,000 real electricity complaints, labelled into four severity tiers; validated experimentally against a traditional manual workflow with two 30-person teams.

Main findings: 81% faster average response time, 68.8% faster resolution time, workload balance index improved from 0.52 to 0.86, fake-complaint detection accuracy of 92%, user satisfaction rose from 3.1 to 4.6 out of 5.

Advantages: The strongest and most quantitatively validated system in this survey; genuinely integrates urgency prediction, duplicate/hotspot clustering, and workload balancing in one pipeline; explicit fake-complaint filtering.

Limitations: Domain-specific to electricity complaints in its case study (though the authors claim generalisability); severity/urgency is a black-box neural-network output, not an explainable score a citizen can see the reasoning behind; no public-facing transparency layer — this is entirely an internal administrative optimisation tool.

Research gap: Extremely strong on internal efficiency, but has no citizen-facing public transparency or registry at all, and its urgency label is not explainable to the person who filed the complaint.

Relation to our project: VidyutSaathi is the closest paper in this survey to UrbanFix's "severity-aware triage" objective, but it solves it with an opaque multi-task neural network aimed at internal department efficiency, whereas UrbanFix solves the same problem with a transparent weighted questionnaire aimed at citizen-visible, explainable prioritisation. UrbanFix does not yet attempt VidyutSaathi's duplicate-clustering or workload-balancing capabilities.

ReGenX — AI-Driven Smart Waste Management with Geofencing and Civic Accountability [4]

What it is about: ReGenX is a mobile-app platform (Resident, Collector, Official roles) for municipal waste management that notifies residents when a collection vehicle is nearby and lets them file waste-related complaints.

Problem addressed: Existing AI/IoT waste-management systems depend on costly smart bins and sensors, which are a "dealbreaker" for cost-sensitive cities, and none of them build in direct citizen feedback or real-time accountability of collection vehicles.

Objective: To deliver geofencing-based collection alerts and AI-assisted complaint handling without expensive IoT hardware.

Method/Approach: Collector GPS positions are tracked via FusedLocationProviderClient and compared to resident geofence radii using the Haversine great-circle-distance formula; Firebase Cloud Messaging pushes real-time "truck nearby" alerts; residents can file complaints (missed collection, overflow, delayed pickup) that are auto-classified by an on-device TensorFlow-Lite NLP model; a separate lightweight on-device TFLite image classifier sorts waste photos into biodegradable/recyclable/hazardous/other categories.

Technology/Algorithms: Haversine distance geofencing, Firebase Realtime Database/Auth/Cloud Messaging, TensorFlow Lite (on-device NLP and image classification), Kotlin/Android.

Dataset/Experiment: 2,000 labelled waste images for the image-classification model; 1,500 labelled complaint texts for the NLP grievance classifier; field-tested across three test localities.

Main findings: 85.2% waste-image classification accuracy at ~95 ms/image on-device; 96.4% geofence-notification success rate; 90.1% grievance-classification accuracy with a 75% cut in manual sorting time.

Advantages: Genuinely low-cost and hardware-free (no IoT sensors needed); on-device AI keeps data private and works even with weak connectivity; combines geofencing, grievance NLP, and image classification in one lightweight mobile stack.

Limitations: Domain-specific to waste collection, not general civic infrastructure complaints; depends on Android/mobile app installation (no account-less web option); citizen visibility is limited to a resident's own complaints and nearby-truck alerts — no public, all-complaints registry; still relies on Firebase (a cloud dependency) despite the on-device inference claim.

Research gap: Solves the cost-of-IoT-hardware gap well, but citizen-facing transparency and cross-department applicability beyond waste are left unaddressed.

Relation to our project: ReGenX's cost-consciousness (avoiding expensive infrastructure) echoes UrbanFix's own choice of a lean, open-source MERN-style stack instead of GPU/IoT hardware, but ReGenX is scoped to waste collection alone, while UrbanFix targets ten general infrastructure categories and a citizen-visible registry that ReGenX does not have.

CivicMate — An AI-Driven City Grievances [5]

What it is about: CivicMate is a web-based grievance platform where citizens upload a photo and short description, get automatic location detection, and receive SMS status updates; it also has a community view with upvoting.

Problem addressed: Existing portals like CPGRAMS lack real-time tracking; citizens are also uncertain their complaint is even authentic or reaches the right department, and administrators lack tools to detect recurring problem areas.

Objective: To simplify complaint filing, verify uploaded images are appropriate, auto-categorise the issue, and let citizens see and support (via upvotes) complaints from others nearby.

Method/Approach: After secure login/registration, a citizen submits a description, photo, and auto-detected location; the AI engine runs a NudeNet-based (YOLOv8/RetinaNet-lineage) NSFW image-moderation check before accepting the photo, then predicts a complaint category; an AI chatbot guides first-time users; a municipal dashboard lets staff assign, track, and resolve complaints.

Technology/Algorithms: NudeNet NSFW-detection model (ResNet-101/YOLO backbones), CNN classifiers, MongoDB, AI chatbot.

Dataset/Experiment: NSFW detector trained on roughly 160,000 auto-labelled images; fine-grained classification tested across 6-18 anatomical categories for content moderation (not civic-defect categories).

Main findings: NSFW detection precision 0.98, recall 0.96, F1 0.97; granular classification F1-scores of 0.945-0.950; GPU inference latency ~24 ms, CPU/edge latency ~0.41 s.

Advantages: Requires no manual account creation friction beyond normal signup; community upvoting surfaces high-impact issues; strong, well-benchmarked image-moderation pipeline; chatbot lowers the barrier for first-time users.

Limitations: Its "AI verification" step only screens for inappropriate (NSFW) content — it does not verify that the photo actually shows the claimed civic defect, which is a materially different and harder problem; still requires account registration (no account-less flow); community view surfaces some complaints publicly but there is no single unified public registry with full status history for every complaint filed.

Research gap: Confuses content-safety moderation with defect-authenticity verification — a photo can pass CivicMate's NSFW check while still not showing a real pothole at all.

Relation to our project: CivicMate's upvote-based "community complaints view" is the closest thing in this survey to UrbanFix's public registry, but it is opt-in/social rather than a complete, always-public record of every complaint's full status history. Its NSFW check is a genuinely useful safety feature UrbanFix's proposed YOLO layer does not (and should not) try to replace, since UrbanFix's AI layer targets a different question — "does this photo match the claimed defect category" — not "is this photo appropriate."

CivicX — An AI-Driven Data-Centric Platform for Urban Grievance Management and Intelligent Plan Approval [6]

What it is about: CivicX combines two municipal functions in one platform: citizen grievance redressal (with geospatial heatmaps) and AI-assisted architectural building-plan approval.

Problem addressed: Grievance systems and construction-plan approval usually run as separate departments with no shared data; plan approval in particular is still manual, document-heavy, and inconsistent across reviewers.

Objective: To unify grievance handling and regulatory plan screening in one data-centric ecosystem with predictive analytics.

Method/Approach: Complaint text, images, and GPS metadata are embedded and classified (Random Forest/SVM/transformer options) to the right department and urgency; a Geospatial Analytics Engine builds real-time heatmaps of complaint density; separately, building-plan submissions are checked by deterministic rule validation and then a supervised ML model (XGBoost/Gradient Boosting/Logistic Regression) predicts "Likely Approved / Requires Review / Non-Compliant."

Technology/Algorithms: Node.js/Express, MongoDB, Python ML microservices, WebSocket real-time sync, geospatial (Haversine/PostGIS-style) indexing.

Dataset/Experiment: Evaluated with synthetic and real complaint samples plus a curated set of compliant/borderline/non-compliant building-plan cases; a controlled study benchmarked CivicX against manual officer classification and a baseline logistic-regression model.

Main findings: 91.4% grievance-classification accuracy (vs. 94.1% for manual officer classification and 82.6% for a baseline ML model); plan-approval F1-score of 87.4% (vs. 90.3% manual, 76.8% baseline); throughput of about 100 requests in ~42 seconds on a standard CPU.

Advantages: Unusually honest benchmarking — reports that trained human officers still slightly outperform the AI system, rather than overselling accuracy; combines two normally-separate municipal workflows; strong geospatial heatmap tooling; role-based access control and encryption-at-rest for privacy.

Limitations: Grievance-classification accuracy (91.4%) is still a few points below manual officer classification (94.1%), and the authors' own confusion cases occur in borderline/ambiguous complaints — the same class of case a photo-verification layer like UrbanFix's would also struggle with; the citizen sees status via the app but there is no description of a registry openly showing every citizen's complaint and its history to other citizens; plan-approval and grievance modules, while integrated on the backend, remain functionally separate from the citizen's point of view.

Research gap: A genuinely integrated architecture, but transparency is aimed at officials (dashboards, audit logs) rather than at the general public; no visual-evidence-authenticity check for grievance photos, only classification of complaint text.

Relation to our project: CivicX's honest human-vs-AI benchmarking is a useful methodological lesson for UrbanFix's future AI validation layer: when it is eventually implemented and tested, it should similarly be benchmarked against manual admin review rather than assumed to be superior. CivicX's geospatial heatmap is also a capability UrbanFix does not currently have (UrbanFix's PPT does not describe GPS/geolocation capture at all, unlike [1], [4], and [6]).

Smart City Concern — An AI-Driven Multimodal Civic Grievance Management System [7]

What it is about: Smart City Concern is a multimodal grievance-intelligence platform that ingests complaints both from a web portal and by monitoring civic-related posts on X (formerly Twitter).

Problem addressed: Complaints scattered across a web portal and social media are hard to process manually, especially given informal language, and existing systems do not merge these two channels or check for duplicates across them.

Objective: To classify, extract structured entities from, prioritise, and de-duplicate complaints regardless of whether they arrive through the web portal or social media.

Method/Approach: A zero-shot transformer (facebook/bart-large-mnli) classifies complaint text into one of five departments without needing department-specific retraining; a transformer-based NER pipeline extracts locations, infrastructure keywords, and urgency terms; a DistilBERT sentiment classifier scores polarity from -1 to +1 and buckets it into High/Medium/Low priority; OCR (Tesseract) extracts text embedded in complaint photos (e.g. signboards) and merges it into the complaint text; TF-IDF plus cosine similarity (threshold 0.85) links duplicate complaints to one existing ticket instead of creating new ones.

Technology/Algorithms: HuggingFace Transformers (zero-shot BART, DistilBERT), Tesseract OCR, TF-IDF/cosine similarity, Flask, MongoDB, X (Twitter) API v2.

Dataset/Experiment: 1,000 complaints across five civic departments for classification benchmarking; 1,500 complaints across five classes for the standalone grievance classifier reused from department-routing evaluation.

Main findings: The proposed zero-shot transformer reaches 92.8% classification accuracy, beating Naive Bayes (72%), SVM (79%), Random Forest (84%), and DistilBERT (89%); duplicate detection is about 89% accurate.

Advantages: Zero-shot classification means new department categories can be added without retraining — a genuinely useful flexibility advantage; merges web and social-media channels into one pipeline; OCR lets sign/notice text embedded in a photo contribute to classification; sentiment-based priority avoids needing a labelled urgency dataset.

Limitations: The authors themselves note informal language, sarcasm, or missing context on social media can reduce accuracy; sentiment polarity is used as a proxy for urgency, but the authors admit this "cannot necessarily identify actual urgency" — an angry but low-severity complaint could outrank a calmly worded but genuinely dangerous one; no citizen-facing public registry — the output goes to an official dashboard only; no explicit photo-defect verification, only OCR text extraction from images.

Research gap: Sentiment is used as a stand-in for severity rather than a designed, explainable priority scheme, and — as with every other paper reviewed — there is no publicly visible, all-complaints registry.

Relation to our project: Smart City Concern's sentiment-based priority score is the paper closest in spirit to UrbanFix's priority mechanism, but it is a fundamentally different and more opaque approach: a sentiment classifier scores the emotional tone of the text, while UrbanFix's weighted questionnaire scores structured, category-specific answers the citizen explicitly provides, making the resulting score easier to explain and defend to a citizen who asks "why is my complaint marked low priority?"

7. Literature Matrix

Ref./Paper	Problem	Method/Tech	Main Finding	Limitation	Research Gap
[1] Infra-Alert (2025)	No real-time tracking/geolocation in portals	Geolocation API, Reverse Geocoding, Cloudinary	More features than CPGRAMS/ICMC/Public Eye	GPS/internet dependence; no AI verification (future work)	No automated visual validation or urgency scoring
[2] CitiWatch (2026)	Manual routing; identity exposure discourages reporting	TF-IDF + ML classifier, OTP, JWT	96.8% routing accuracy; 85% faster allocation	No images; no public registry	No visual evidence; no public transparency
[3] VidyutSaathi (2026)	Classification, duplicates, workload treated separately	Sentence-BERT, multi-task NN, DBSCAN	81% faster response; 68.8% faster resolution	Opaque NN urgency score; no public view	No citizen-facing transparency; not explainable
[4] ReGenX (2025)	High-cost IoT hardware for waste management	Haversine geofencing, on-device TFLite	96.4% notification success; 85.2% image accuracy	Waste-domain only; own-complaints view only	Not generalised beyond waste; no public registry
[5] CivicMate (2026)	Inauthentic images; no recurring-issue detection	NudeNet NSFW model, CNN, chatbot	NSFW F1 = 0.97	Verifies safety, not defect authenticity; needs account	Confuses moderation with defect verification
[6] CivicX (2026)	Grievance & plan approval run as separate silos	RF/SVM/transformer, XGBoost, geo-heatmap	91.4% classification vs. 94.1% manual	AI below human accuracy; official-only transparency	No public registry; no photo-authenticity check

Ref./Paper	Problem	Method/Tech	Main Finding	Limitation	Research Gap
[7] Smart City Concern (2026)	Web + social-media complaints processed separately	Zero-shot BART, DistilBERT, OCR, TF-IDF dedup	92.8% classification accuracy	Sentiment $\neq$ true urgency (authors' own admission)	Urgency proxy opaque; no public registry

8. Comparative Analysis

Common Points

- All seven papers target urban civic infrastructure complaints and were built, tested, or piloted in Indian metro or semi-urban college-campus settings (Belagavi, Coimbatore, Pune, Bengaluru, Navi Mumbai, Vasai, Nalgonda) ; none discuss a rural or panchayat-level deployment.
- Every paper except [4] uses some form of NLP-based text classification (TF-IDF, sentiment analysis, zero-shot transformers, or a multi-task neural network) to automatically route or score complaints ; a capability UrbanFix's current MVP does not yet implement (routing is still manual via the admin console).
- Reported classification accuracies cluster in a fairly narrow high-90s to low-90s band ([2] 96.8%, [3] implied high performance, [6] 91.4%, [7] 92.8%), suggesting this specific sub-problem (text-to-department routing) is close to a solved problem in the literature.
- Citizen-facing transparency, where present at all, is limited to the submitter's own complaint status ([1], [4]) or an internal official dashboard ([2], [3], [6], [7]) only [5]'s community upvote view is genuinely visible to other citizens, and even that is opt-in and partial rather than a complete public registry.
- None of the seven papers verify that an uploaded photo actually shows the claimed defect; where image AI is used, it is for storage ([1]), waste-type classification ([4]), or content-safety moderation ([5]) never defect-category authenticity, which is exactly the gap UrbanFix's proposed YOLO layer targets.

Differences

- Authentication/identity model: [2] uses mobile-number OTP for persistent-but-anonymous identity; [5] and [6] use standard account registration; [1], [3], [4], [7] do not foreground citizen authentication at all (complaints are department-facing). None use UrbanFix's specific pattern of a one-time email OTP with no account ever created.
- Priority/urgency mechanism: rule-based severity tiers from a trained multi-task neural network in [3]; sentiment polarity in [7]; no explicit urgency mechanism in [1], [4], [5]; a routing model in [2] and [6] that predicts department but not a citizen-visible urgency label.
- Duplicate detection: present and benchmarked in [3] (spatiotemporal clustering) and [7] (TF-IDF/cosine similarity); absent in [1], [2], [4], [5], [6].
- Domain scope: [4] is scoped specifically to waste collection; [6] additionally covers building-plan approval (a different municipal function entirely); the rest are general multi-category infrastructure complaint systems, as UrbanFix also is.

- Channel of complaint intake: [7] uniquely also monitors social media (X/Twitter) alongside a web portal; the others are single-channel (web or app).

9. Common Findings

Taken together, the seven papers show that AI-based text classification for department routing is a mature, well-benchmarked sub-problem, with several independent teams reporting accuracies above 90% using different techniques (TF-IDF+ML, zero-shot transformers, multi-task neural networks). By contrast, none of the seven papers has solved — or even fully attempted — citizen-facing public transparency in the form of an always-visible, all-complaints registry, and none has attempted photo-authenticity verification of the reported defect itself (as opposed to content moderation or waste-type sorting). This is the specific combination of gaps that motivates UrbanFix.

10. Research Gaps

Explicit Gaps (stated by the papers' own authors)

- [1] Infra-Alert explicitly lists "AI-driven complaint verification" and "automated prioritisation" as future enhancements, not implemented in the current system.
- [4] ReGenX's authors note the prototype has no real-time IoT sensor feed for actual bin fill-levels and needs wider stress-testing before large-scale deployment.
- [6] CivicX's authors report that its grievance-classification model (91.4%) still trails manual officer classification (94.1%), and identify borderline/ambiguous complaints as the main source of error.
- [7] Smart City Concern's authors state directly that their similarity- and sentiment-based approach "cannot necessarily identify semantic similarity or actual urgency."

Common Limitations (recurring across multiple papers)

- Dependence on stable internet/GPS for geolocation-based features ([1], [4]).
- Transparency limited to the submitter or to administrators — no paper offers an always-public, all-complaints registry with full status history visible to any visitor ([1], [2], [3], [6], [7]; [5] partially, via opt-in community upvoting only).
- Where a priority/urgency mechanism exists, it is an opaque ML output (a neural-network severity class in [3], a sentiment score in [7]) rather than a score whose reasoning a citizen can inspect.
- No paper implements or even proposes a category-specific visual check that the photo genuinely matches the claimed defect — existing image AI use is limited to storage, waste-type sorting, or NSFW moderation ([1], [4], [5]).

Comparison-Based Gaps (visible only when comparing across papers and against UrbanFix)

- No single reviewed paper combines all three of (a) zero-friction, account-less citizen submission, (b) a fully public, all-complaints status registry, and (c) a human in the loop, category-specific photo authenticity check ; each paper offers at most one or two of these three, which is the specific combination UrbanFix is attempting.

- All seven papers are validated in urban or semi-urban metro/college-town settings; none discuss whether their approach would still work where connectivity, smartphone access, or digital literacy is lower this is examined further in Section 11.

11. Under-Explored Problem: The Urban-Rural Gap in Civic Grievance Systems

One part of this domain is under-explored not completely untouched, but clearly under-explored by the literature reviewed here: none of the seven papers test, mention, or design for rural or panchayat level grievance redressal. All seven are deployed, piloted, or demonstrated in urban or semi-urban Indian settings Belagavi, Coimbatore, Pune, Bengaluru, Navi Mumbai, Vasai, and Nalgonda are the specific locations named across [1]-[7] and all seven implicitly assume a citizen who has reliable smartphone/internet access, is comfortable with a web or mobile-app interface, and can take and upload a clear photo. This is called "under explored" rather than "untouched" because rural digital-governance research exists more broadly outside this specific set of seven papers; but within the literature that is most directly comparable to UrbanFix, this gap is real and consistent across all seven.

This matters because rural civic issues hand pump failures, unpaved roads, drainage in villages are just as much in need of transparent tracking and public accountability as urban potholes and streetlights, but a system tuned for smartphone literate urban users (multi-step forms, high resolution photo upload, app installation as in [4] and [5]) may simply not transfer to a rural citizen with a basic phone, patchy connectivity, or lower comfort with English-language digital forms.

UrbanFix's current design takes a modest step in this direction rather than solving it outright: its account-less, email-OTP-only flow removes the friction of app installation and password management that [4] and [5] require, and its ten fixed categories with a short weighted questionnaire are simpler to complete than a free-text description. However, it is important to be honest that UrbanFix still assumes web access and a working email address, so it does not yet fully close the urban-rural gap — this remains realistic future scope (Section 15) rather than a solved problem, and a genuine test of UrbanFix in a semi-urban or rural pilot would be needed to support any stronger claim.

12. Proposed Project in Relation to Existing Research

Research Gap	Shown in	Existing Limitation	UrbanFix Feature	How UrbanFix Attempts to Address It
No public, all-complaints registry	[1],[2],[3],[6],[7]	Status visible only to the submitter or to admins	Public complaint registry with full status history	Existing studies mainly focus on admin-facing tracking, while our project focuses more on making every complaint's history visible to any visitor, not just the filer.
Urgency score is opaque (ML/sentiment)	[3],[7]	Neural-network severity class or sentiment polarity used as urgency, not explainable to the citizen	Weighted-questionnaire priority score	Our project attempts to address this limitation by using a fixed, inspectable set of questionnaire weights, so a citizen can see which of their answers drove the score, rather than a numeric ML output.

Research Gap	Shown in	Existing Limitation	UrbanFix Feature	How UrbanFix Attempts to Address It
No photo-authenticity verification	[1],[4],[5],[6]	Image AI limited to storage, waste-sorting, or NSFW moderation — never checks the photo matches the claimed defect	Proposed per-category YOLO validation layer with confidence-based routing	The proposed system aims to route low-confidence photo-category mismatches to a human admin review queue rather than auto-publishing them, keeping accountability at every decision point.
Account-creation friction discourages reporting	[5],[6]	Standard registration/login required before a complaint can be filed	OTP-email-only, account-less submission	Compared with the reviewed approaches, our project removes account creation entirely, verifying only the current submission rather than creating a persistent user identity.
No system tested for rural/low-resource contexts	[1]-[7] (all)	All validated in urban/semi-urban metro or campus settings	Account-less, simple structured-form design	This is only a partial step: the project provides an approach that reduces some friction, but genuinely closing this gap would need dedicated rural piloting, which is left to future scope.

13. Key Advantages of the Proposed Project

No-account, OTP-only submission: Removes the registration/login step that [5] and [6] both require, which prior research (and common sense) links to reporting friction and drop-off before a complaint is even filed. This directly relates to the accessibility gap named in the project's own objectives.

Explainable, weighted-questionnaire priority score: Unlike the opaque neural-network severity output in [3] or the sentiment score in [7] which its own authors admit is not a reliable proxy for actual urgency a citizen can see which of their questionnaire answers contributed to the score, making the triage step auditable rather than a black box.

Public complaint registry with full status history: No other paper reviewed here provides an always-visible, all-complaints record open to any visitor; [1]'s dashboard and [2]/[3]/[6]/[7]'s admin dashboards are not public, and [5]'s community view is opt-in and partial. This is the clearest and most consistent gap this survey found.

Human-in-the-loop AI validation design (proposed): Even once the YOLO validation layer is built, low-confidence detections are routed to an admin review queue rather than auto-published or auto-rejected a safeguard that CivicX's [6] honest human-vs-AI benchmarking (officers still outperform the model) suggests is a sensible design choice rather than an unnecessary caution.

JWT-protected admin console: Gives UrbanFix the same baseline access-control discipline that [2] and [6] apply to their own dashboards, preventing unauthorised status changes.

PDF receipt as a tangible proof-of-submission artifact: None of the seven reviewed papers describe issuing the citizen a downloadable, portable proof of their complaint; this is a distinctive addition to UrbanFix's workflow.

Lean, open-source, low-infrastructure stack: Echoes [4]'s cost-conscious argument against expensive IoT/GPU infrastructure, but applies the same philosophy to complaint tracking rather than waste-bin sensors the whole system runs on React/Express/MongoDB with no specialised hardware.

14. Key Value / Contribution

Problem value: It addresses the four-part systemic breakdown named in the project's own problem statement no tracking ID, no public record, no urgency signal, no photo verification that is present, in some combination, across every paper reviewed.

User value: Citizens benefit from a lower-friction, account-less way to report issues and gain the ability to see any complaint's status, not just their own; municipal admins benefit from a structured priority signal and a JWT-secured console.

Technical value: A working end to end MERN-style stack (React/Vite, Express/Node.js, MongoDB with TTL-indexed OTPs) plus a clearly architected extension point for per-category YOLO photo validation with confidence-based routing.

Practical value: The system is deployable without specialised hardware, in line with the cost-consciousness that [4] argues is essential for real-world municipal adoption in developing urban contexts.

Research value: It targets the specific combination of gaps public transparency, explainable triage, and human-in-the-loop visual validation that this survey found no single reviewed paper addresses together.

Innovation value: The main point of difference is not any single technique (each individual idea OTP verification, weighted scoring, photo AI has some precedent in the literature) but the specific combination of a fully public registry with an explainable score and a human-in-the-loop AI layer, which this survey did not find combined elsewhere.

15. Existing Research vs. Our Proposed Project

Aspect	Existing Research ([1]-[7])	UrbanFix (Proposed/Implemented)
Main Problem	Manual routing, weak tracking, fragmented channels	Fragmented, untracked, invisible civic reporting
Target Users	Citizens + municipal officials, mostly urban/campus contexts	Citizens + municipal admin; account-less citizen access
Authentication	Standard login ([5],[6]) or mobile-OTP identity ([2])	Email OTP only, no account created
Complaint Routing	Automated ML/NLP routing in [2],[3],[6],[7]	Manual, via admin console (not yet automated)
Priority Mechanism	ML severity class [3]; sentiment score [7]; none in [1],[4],[5]	Explainable weighted questionnaire, computed live
Photo Handling	Storage [1]; waste-type sort [4]; NSFW filter [5]; none in [2],[3],[6],[7]	Proposed category-specific YOLO defect-authenticity check
Transparency	Admin/self-only in [1],[2],[3],[6],[7]; partial community view in [5]	Fully public registry, full status history, any visitor
Rural Readiness	Not addressed in any of the seven papers	Not solved, but lower-friction design; explicit future scope
Deployment Cost	Ranges from low ([4]) to moderate (ML	Low — open-source MERN-style stack, no

Aspect	Existing Research ([1]-[7])	UrbanFix (Proposed/Implemented)
	microservices in [3],[6])	GPU/IoT hardware

16. Limitations

Limitations of Existing Research

- Geolocation and connectivity-dependent features degrade in poor-network or indoor conditions ([1], [4]).
- Where urgency/priority is computed at all, it is an opaque ML output rather than something a citizen can inspect the reasoning behind ([3], [7]).
- No paper offers a fully public, all complaints registry transparency, where present, stops at the submitter or the administrator.
- AI-based photo handling is limited to storage, content moderation, or waste-sorting none attempt category-specific defect verification, so this remains an open technical problem, not something this survey can say has already been solved elsewhere.
- Several reported accuracy figures ([2] 96.8%, [7] 92.8%) come from datasets curated or generated by the same research team, so real-world performance on messier, more informal citizen input (as [6] and [7] themselves note) is likely somewhat lower.

Possible Limitations of Our Project

- The AI photo-validation layer is currently a proposed architecture only not yet implemented, trained, or tested so no accuracy claim can be made for it yet.
- The system is web-only; there is no native mobile app, unlike [4] and [5], which may matter for citizens who are more comfortable with mobile apps than mobile web.
- Complaint routing to the correct department is still manual via the admin console, unlike the automated ML routing demonstrated in [2], [3], [6], and [7].
- No duplicate-complaint detection exists yet, unlike [3] and [7], so the same pothole reported by multiple citizens would currently appear as separate entries in the public registry.
- The weighted questionnaire's specific weights are a design choice by the team rather than something learned from data or validated against real complaint outcomes, so their real-world accuracy as an urgency proxy is currently untested.
- Email-OTP-only verification may exclude citizens without regular email access, which is a relevant consideration for the urban-rural gap discussed in Section 11.
- No multilingual support and no analytics/reporting dashboard are currently built, both of which are explicitly listed as out-of-scope in the project's own scope slide.
- The system has not yet been pilot-tested with real citizens or a real municipal body, so all of the above are architectural/design limitations rather than measured ones.

17. Future Scope

- Implement and benchmark the proposed per-category YOLO photo-validation layer, and following CivicX's [6] example report its accuracy honestly against manual admin review rather than assuming it will outperform humans.
- Add geolocation/GPS capture and a geospatial heatmap of complaint density, similar to [1], [4], and [6], to help authorities spot recurring problem zones.
- Automate department routing using an NLP classifier (zero-shot, as in [7], would be a good fit given it needs no retraining as new categories are added) to reduce reliance on manual admin triage.
- Add duplicate/near-duplicate complaint detection, following [3]'s spatiotemporal clustering or [7]'s TF-IDF/cosine-similarity approach, so the same issue reported by multiple citizens is merged into one registry entry.
- Build a native mobile application, as in [4] and [5], to widen access beyond web OR mobile-browser users.
- Add multilingual support, which is listed as future scope in the project's own slides and is also relevant to closing the urban-rural accessibility gap discussed in Section 11.
- Conduct a genuine pilot in a semi-urban or rural setting to test whether UrbanFix's account-less, low-friction design actually improves access for lower-connectivity users, rather than assuming it does.
- Add an analytics/reporting dashboard for municipal officials, as is present in some form in [3] and [6], once enough real complaint data has been collected.

18. Conclusion

This survey reviewed seven IEEE papers on AI-assisted civic grievance management and compared them against UrbanFix. A consistent pattern emerged: text-based complaint classification is a fairly mature sub-problem, with several independent systems reporting classification accuracies above 90%, but none of the seven papers provides a fully public, all-complaints registry with visible status history, none offers a citizen-inspectable, explainable priority score, and none attempts to verify that an uploaded photo genuinely matches its claimed defect category rather than merely storing it, sorting it, or moderating it for inappropriate content. UrbanFix's account-less OTP flow, weighted-questionnaire priority score, and public registry are a direct response to this specific combination of gaps, and its proposed human-in-the-loop YOLO validation layer is designed to fill the remaining photo-verification gap without removing human accountability. At the same time, the survey also surfaced a gap that UrbanFix has not fully closed: none of the seven papers, and UrbanFix itself, have been tested outside urban or semi-urban settings, so the urban-rural accessibility question remains genuinely open rather than solved.

19. References

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Appendix: Citation Verification Table

Citation	Paper	Information Used
[1]	Infra-Alert	Problem/method/findings/limitations summarised in Section 6; used in comparative matrix (Sec. 7), gap analysis (Sec. 10), and gap-to-feature mapping (Sec. 12).
[2]	CitiWatch	Problem/method/findings/limitations summarised in Section 6; used in comparative matrix (Sec. 7), gap analysis (Sec. 10), and gap-to-feature mapping (Sec. 12).
[3]	VidyutSaathi	Problem/method/findings/limitations summarised in Section 6; used in comparative matrix (Sec. 7), gap analysis (Sec. 10), and gap-to-feature mapping (Sec. 12).
[4]	ReGenX	Problem/method/findings/limitations summarised in Section 6; used in comparative matrix (Sec. 7), gap analysis (Sec. 10), and gap-to-feature mapping (Sec. 12).
[5]	CivicMate	Problem/method/findings/limitations summarised in Section 6; used in comparative matrix (Sec. 7), gap analysis (Sec. 10), and gap-to-feature mapping (Sec. 12).
[6]	CivicX	Problem/method/findings/limitations summarised in Section 6; used in comparative matrix (Sec. 7), gap analysis (Sec. 10), and gap-to-feature mapping (Sec. 12).
[7]	Smart City Concern	Problem/method/findings/limitations summarised in Section 6; used in comparative matrix (Sec. 7), gap analysis (Sec. 10), and gap-to-feature mapping (Sec. 12).